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FINALTERM EXAMINATION

SEMESTER FALL 2004

MTH501 – Linear Algebra

Total Marks: 65

Duration:120mins

StudentID/LoginID	
Name	
PVC Name/Code	
Date	

Maximum Time Allowed: (2 Hour)

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.

a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.

b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.

c. Write all steps, missing steps may lead to deduction of marks.

****WARNING: Please note that Virtual University takes serious note of unfair means. Anyone found involved in cheating will get an `F` grade in this course.**

For Teacher's use only

Question	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Total
Marks													

Question No: 1 **Marks: 2**

Consider a system of linear equations A x =b where A is a 3×3 matrix having 3 pivot positions, then which statement is false about the system Ax =b

(a) System has unique solution.
(b) Rank of the matrix is 3.
(c) There is only one free variable in solution of that system.
(d) The associated homogeneous system Ax =0 has only trivial system.

Question No: 2 **Marks: 2**

If a finite set S of non zero vectors span a vector space V, then some subset of S is a basis for V.

1. True
2. false

Question No: 3 **Marks:2**

Angle between the vectors $x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ & $y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ is,

(a) $\frac{\pi}{2}$

(b) π

(c) 0

(d) $\frac{\pi}{3}$

Question No: 4 **Marks:2**

If λ is an eigenvalue of the invertible matrix A then an eigenvalue of A^{-1} will be

(a) λ^{-1}

(b) $\frac{1}{\lambda^2}$

(c) λ^2

(d) Can't be determined.

Question No: 5 **Marks:2**

If rank of a 3×5 matrix is 3 then dimension of its Null space is

(a) 0
(b) 3
(c) 2
(d) We can't say anything.

Question No: 6 **Marks:5**

Find all 2×2 matrices for which $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is an eigenvector correspond to the eigenvalue 5.

Question No: 7 **Marks:10**

Compute the determinant $\begin{vmatrix} 6 & 3 & 2 & 4 & 0 \\ 9 & 0 & -4 & 1 & 0 \\ 8 & -5 & 6 & 7 & 1 \\ 3 & 0 & 0 & 0 & 0 \\ 4 & 2 & 3 & 2 & 0 \end{vmatrix}$.

Question No: 8 **Marks:10**

Determine whether the matrix $A=\begin{bmatrix}1 & 0 & 1\\ 0 & 1 & 0\\ 1 & 0 & 1\end{bmatrix}$ is diagonalizable or not?

Question No: 9 **Marks:5**

Consider the Linear transformation defined by T(x) = Ax where $A=\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\\ 7 & 8 & 9\end{bmatrix}$, then find $T\begin{bmatrix}1\\0\\0\end{bmatrix}$ *and* $T\begin{bmatrix}1\\1\\0\end{bmatrix}$.

Question No: 10 **Marks:10**

Find the B-matrix of the transformation $x \to Ax$ *where* $A=\begin{bmatrix}-14 & 4 & -14\\ -33 & 9 & -31\\ 11 & -4 & 11\end{bmatrix}$, when B= {$b_1$, b_2, b_3}

where $b_1=\begin{bmatrix}-1\\-2\\1\end{bmatrix}, b_2=\begin{bmatrix}-1\\-1\\1\end{bmatrix}, b_3=\begin{bmatrix}-1\\-2\\0\end{bmatrix}$.

Question No:11 **Marks: 5**

For the bases $\beta=\{2x^2-x, 3x^2+1, x^2\}$ *of* P_2 find the coordinate of the $6x^2-x+1$ relative to β.

Question No:12 **Marks:10**

Find a least squares solution of the inconsistent system Ax =b where $A=\begin{bmatrix}1 & 1\\ 1 & 2\\ 1 & 3\end{bmatrix}, b=\begin{bmatrix}0\\0\\6\end{bmatrix}$.